CSAI2121 Probability and Statistics

Midterm Revision Handout

Practice questions and conceptual review

Name: _____ **Date:** _____

How to use this handout. First review the concepts below. Then try the practice questions without looking at the solutions. These questions cover similar skills to those assessed in the midterm test, but they are not copies of the test questions.

Part I. Key Concepts

1. Counting and probability rules

- If order does *not* matter, use combinations:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

- A useful counting strategy is

$$\text{valid outcomes} = \text{all outcomes} - \text{invalid outcomes}.$$

- For events A and B,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

- Conditional probability and Bayes' theorem:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0.$$

  Always check which event is the *condition*.

2. Discrete random variables

- **Binomial:** Count successes in a fixed number of independent trials with the same success probability.
- **Geometric:** Count failures before the first success. In this handout, its possible values are $0, 1, 2, \ldots$.
- **Poisson:** Model a count of occurrences over a specified interval when a Poisson model is assumed.
- To calculate $P(X \geq 1)$, consider the complement:

$$P(X \geq 1) = 1 - P(X = 0).$$

3. Continuous random variables

- A PDF must be nonnegative and satisfy

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

- The CDF is

$$F_X(x) = P(X \le x) = \int_{-\infty}^{x} f_X(t)\,dt.$$

  State its value *outside* the support as well.

- If X is continuous, $P(X = a) = 0$ for every individual value a.

- For a normal random variable,

$$Z = \frac{X - \mu}{\sigma}.$$

  Check carefully whether the question asks for an upper-tail or lower-tail probability.

4. Joint distributions and independence

- Obtain a marginal distribution by summing or integrating out the other variable.

- Discrete independence requires

$$p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

  for *every* possible pair (x, y).

- Continuous independence requires

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

  almost everywhere.

- Conditional independence given a third variable does *not* necessarily imply independence without that condition.

Common mistakes to avoid: confusing $P(A \mid B)$ with $P(B \mid A)$; forgetting to subtract an intersection in the addition rule; treating a PDF value as a probability; and assuming that equal-looking marginals imply independence.

Part II. Multiple-Choice Practice

Choose the one best answer for each question.

1. A model designer selects three distinct variables from seven candidates. How many selections are possible if order does not matter?

   (a) 21
   (b) 35
   (c) 210
   (d) 343

2. Suppose $P(A) = 0.4$, $P(B) = 0.3$, and $P(A \cap B) = 0.12$. What is $P(A \cup B)$?

   (a) 0.12
   (b) 0.48
   (c) 0.58
   (d) 0.70

3. Let G be the number of failures before the first success, with success probability 0.2 on each independent trial. What is $P(G = 2)$?

   (a) 0.04
   (b) 0.128
   (c) 0.16
   (d) 0.512

4. A box contains eight components, of which three are marked. Two components are selected without replacement. What is the probability that both selected components are marked?

   (a) $\dfrac{3}{8}$
   (b) $\dfrac{3}{28}$
   (c) $\dfrac{9}{64}$
   (d) $\dfrac{1}{28}$

5. A component is defective with probability 0.1. A test returns a positive result with probability 0.9 for a defective component and probability 0.2 for a non-defective component. What is the probability that a component is defective given a positive result?

   (a) $\frac{1}{9}$
   (b) $\frac{1}{4}$
   (c) $\frac{1}{3}$
   (d) $\frac{9}{10}$

6. A random variable has PDF $f_X(x) = 3x^2$ for $0 \leq x \leq 1$, and zero otherwise. What is $P(X \leq \frac{1}{2})$?

   (a) $\frac{1}{16}$
   (b) $\frac{1}{8}$
   (c) $\frac{1}{4}$
   (d) $\frac{1}{2}$

7. Suppose $X \sim N(20, 3^2)$ and $\Phi(1) \approx 0.8413$. What is $P(X > 23)$?

   (a) 0.8413
   (b) 0.1587
   (c) 0.5000
   (d) 0.9772

8. The joint PMF of U and V is

	$V = 0$	$V = 1$
$U = 0$	0.1	0.4
$U = 1$	0.2	0.3

   What is $P(U = 1 \mid V = 0)$?

   (a) $\frac{1}{5}$
   (b) $\frac{1}{3}$
   (c) $\frac{1}{2}$
   (d) $\frac{2}{3}$

9. Which statement must hold for every CDF $F_X(x)$?

   (a) F_X is strictly increasing everywhere.
   (b) F_X is nondecreasing.
   (c) F_X has a derivative at every point.
   (d) $F_X(0) = \frac{1}{2}$.

10. If $N \sim \text{Poisson}(1)$, what is $P(N \geq 1)$?

    (a) e^{-1}
    (b) $\dfrac{1}{2}$
    (c) $1 - e^{-1}$
    (d) e^{-2}

Part III. Structured Practice

Show your main steps. Exact answers are preferred where convenient.

11. **Counting and Bayes' Theorem**

(a) Four variables are to be selected from eight candidates. Two particular variables are labelled A and B. How many selections contain *exactly one* of A and B?

(b) A machine produces a faulty item with probability 0.15. A detector signals "fault" with probability 0.90 for a faulty item and probability 0.10 for a non-faulty item. Find:

(i) the overall probability of a "fault" signal;

(ii) the probability that an item is faulty given a "fault" signal.

12. **Discrete Models**

Each of five independently tested inputs causes a program to fail with probability 0.4. Let X be the number of failures.

(a) State the distribution of X and write its PMF.

(b) Find $P(X \leq 1)$.

(c) Find $E[X]$ and $\mathrm{Var}(X)$.

(d) Separately, let $Y \sim \mathrm{Poisson}(1.5)$. Express $P(Y \geq 2)$ in terms of $e^{-1.5}$.

13. **Continuous Models**

The PDF of a random variable R is

$$f_R(r) = \begin{cases} cr, & 0 \le r \le 3, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find c.

(b) Find $F_R(r)$ for all real values of r.

(c) Find $P(1 < R < 2)$.

(d) Find $E[R]$ and $\text{Var}(R)$.

(e) Separately, if $W \sim N(40, 4^2)$, find $P(W < 44)$ using $\Phi(1) \approx 0.8413$.

(f) If $T \sim$ Exponential(3), where 3 is the rate, find

$$P\left(T > \frac{\ln 2}{3}\right).$$

14. **Joint Distributions**

Discrete case. The joint PMF of U and V is

	$V=0$	$V=1$
$U=0$	0.18	0.12
$U=1$	0.42	0.28

(a) Find the marginal PMFs of U and V.

(b) Find $P(U=1 \mid V=1)$.

(c) Are U and V independent? Support your answer with a calculation.

Continuous case. Let

$$f_{R,S}(r,s) = \begin{cases} k(r+s), & 0 \le r \le 1, \quad 0 \le s \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

(d) Find k.

(e) Find $f_R(r)$ and $f_S(s)$, including their supports.

(f) Find $P(R \le \frac{1}{2},\ S \le \frac{1}{2})$.

(g) Are R and S independent? Justify your answer.

Part IV. Integrated Practice

15. **Two Indicators for Priority Events**

An event is classified as high priority (H) with probability

$$P(H) = \frac{1}{3}.$$

Two binary indicators, A and B, are recorded. A value of 1 means the indicator is active. Their conditional probabilities are

	$P(A = 1 \mid \cdot)$	$P(B = 1 \mid \cdot)$
H	$\frac{2}{3}$	$\frac{3}{4}$
H^c	$\frac{1}{6}$	$\frac{1}{2}$

Assume A and B are **conditionally independent given** H, and also conditionally independent given H^c.

(a) Find $P(A = 1)$ and $P(B = 1)$.

(b) Find $P(A = 1, B = 1)$. Show the contribution from each event class.

(c) Find $P(H \mid A = 1)$.

(d) Find $P(H \mid A = 1, B = 1)$.

(e) Find $P(B = 1 \mid A = 1)$.

(f) Are A and B independent *without conditioning on the event class*? Explain.

Reflection questions for discussion

1. Why can $P(H \mid A = 1, B = 1)$ be larger than $P(H \mid A = 1)$?
2. Why does conditional independence not settle whether A and B are independent overall?
3. In a Bayes calculation, what belongs in the denominator?

Teacher's Solutions

Remove before distributing the student handout

Part II. Multiple-Choice Answers

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	B	B	C	B	B	D	B	C

1. $\binom{7}{3} = 35$.
2. $P(A \cup B) = 0.4 + 0.3 - 0.12 = 0.58$.
3. $P(G = 2) = (0.8)^2(0.2) = 0.128$.
4. $P(\text{both marked}) = \dfrac{\binom{3}{2}}{\binom{8}{2}} = \dfrac{3}{28}$.
5. $$P(\text{defective} \mid +) = \frac{0.1(0.9)}{0.1(0.9) + 0.9(0.2)} = \frac{0.09}{0.27} = \frac{1}{3}.$$
6. $P(X \le \frac{1}{2}) = \displaystyle\int_0^{1/2} 3x^2\,dx = \left(\tfrac{1}{2}\right)^3 = \frac{1}{8}$.
7. $P(X > 23) = P\left(Z > \dfrac{23 - 20}{3}\right) = 1 - \Phi(1) = 0.1587$.
8. $P(U = 1 \mid V = 0) = \dfrac{0.2}{0.1 + 0.2} = \dfrac{2}{3}$.
9. Every CDF is nondecreasing.
10. $P(N \ge 1) = 1 - P(N = 0) = 1 - e^{-1}$.

Part III. Structured Practice

Question 11: Counting and Bayes' Theorem

(a) Choose which one of A and B is selected; then choose three of the other six variables:

$$2\binom{6}{3} = 2(20) = \boxed{40}.$$

(b) Let F mean "faulty" and $+$ mean "fault signal". Then

$$P(+) = P(+ \mid F)P(F) + P(+ \mid F^c)P(F^c) = 0.90(0.15) + 0.10(0.85) = \boxed{0.22}.$$

Hence

$$P(F \mid +) = \frac{P(+ \mid F)P(F)}{P(+)} = \frac{0.90(0.15)}{0.22} = \frac{0.135}{0.22} = \boxed{\frac{27}{44}}.$$

Question 12: Discrete Models

(a)

$$X \sim \text{Binomial}(5, 0.4), \qquad P(X = j) = \binom{5}{j}(0.4)^j(0.6)^{5-j}, \quad j = 0, \ldots, 5.$$

(b)

$$\begin{aligned} P(X \leq 1) &= P(X = 0) + P(X = 1) \\ &= (0.6)^5 + 5(0.4)(0.6)^4 \\ &= 0.07776 + 0.25920 \\ &= \boxed{0.33696}. \end{aligned}$$

(c)

$$E[X] = 5(0.4) = \boxed{2}, \qquad \text{Var}(X) = 5(0.4)(0.6) = \boxed{1.2}.$$

(d)

$$P(Y \geq 2) = 1 - P(Y = 0) - P(Y = 1) = \boxed{1 - 2.5e^{-1.5}}.$$

Question 13: Continuous Models

(a) Normalisation gives

$$1 = \int_0^3 cr\, dr = \frac{9c}{2}, \qquad \boxed{c = \frac{2}{9}}.$$

(b)

$$F_R(r) = \begin{cases} 0, & r < 0, \\ \dfrac{r^2}{9}, & 0 \leq r \leq 3, \\ 1, & r > 3. \end{cases}$$

(c)

$$P(1 < R < 2) = F_R(2) - F_R(1) = \frac{4}{9} - \frac{1}{9} = \boxed{\frac{1}{3}}.$$

(d)

$$E[R] = \int_0^3 r\left(\frac{2r}{9}\right) dr = \boxed{2},$$

$$E[R^2] = \int_0^3 r^2\left(\frac{2r}{9}\right) dr = \frac{9}{2}.$$

Therefore,

$$\text{Var}(R) = E[R^2] - \big(E[R]\big)^2 = \frac{9}{2} - 4 = \boxed{\frac{1}{2}}.$$

(e)

$$P(W < 44) = P\left(Z < \frac{44 - 40}{4}\right) = \Phi(1) = \boxed{0.8413}.$$

(f) For an exponential random variable with rate 3, $P(T > t) = e^{-3t}$ for $t \geq 0$. Thus

$$P\left(T > \frac{\ln 2}{3}\right) = e^{-3(\ln 2/3)} = e^{-\ln 2} = \boxed{\frac{1}{2}}.$$

Question 14: Joint Distributions

(a) Sum across rows and columns:

$$p_U(0) = 0.30, \quad p_U(1) = 0.70; \qquad p_V(0) = 0.60, \quad p_V(1) = 0.40.$$

(b)

$$P(U = 1 \mid V = 1) = \frac{0.28}{0.40} = \boxed{0.70}.$$

(c) U and V **are independent**. For example, the four joint probabilities equal the corresponding products of marginals:

$$\begin{aligned} 0.18 &= (0.30)(0.60), \quad 0.12 = (0.30)(0.40), \\ 0.42 &= (0.70)(0.60), \quad 0.28 = (0.70)(0.40). \end{aligned}$$

(d)

$$1 = \int_0^1 \int_0^1 k(r+s)\, ds\, dr = k, \qquad \boxed{k = 1}.$$

(e) For $0 \leq r \leq 1$,

$$f_R(r) = \int_0^1 (r+s)\, ds = r + \frac{1}{2}.$$

For $0 \leq s \leq 1$,

$$f_S(s) = \int_0^1 (r+s)\, dr = s + \frac{1}{2}.$$

Each marginal PDF is zero outside $[0, 1]$.

(f)

$$P\left(R \leq \frac{1}{2}, S \leq \frac{1}{2}\right) = \int_0^{1/2} \int_0^{1/2} (r+s)\, ds\, dr = \boxed{\frac{1}{8}}.$$

(g) R and S **are not independent**. For instance, at $r = s = \frac{1}{4}$,

$$f_{R,S}\left(\tfrac{1}{4}, \tfrac{1}{4}\right) = \frac{1}{2}, \qquad f_R\left(\tfrac{1}{4}\right) f_S\left(\tfrac{1}{4}\right) = \frac{3}{4} \cdot \frac{3}{4} = \frac{9}{16}.$$

The two values differ.

Part IV. Integrated Practice

Question 15: Two Indicators for Priority Events

(a) By total probability,

$$P(A=1) = \frac{1}{3}\cdot\frac{2}{3} + \frac{2}{3}\cdot\frac{1}{6} = \boxed{\frac{1}{3}},$$

$$P(B=1) = \frac{1}{3}\cdot\frac{3}{4} + \frac{2}{3}\cdot\frac{1}{2} = \boxed{\frac{7}{12}}.$$

(b) Use conditional independence *within each event class*:

$$\begin{aligned} P(A=1,B=1) &= P(H)P(A=1\mid H)P(B=1\mid H) \\ &\quad + P(H^c)P(A=1\mid H^c)P(B=1\mid H^c) \\ &= \frac{1}{3}\cdot\frac{2}{3}\cdot\frac{3}{4} + \frac{2}{3}\cdot\frac{1}{6}\cdot\frac{1}{2} \\ &= \frac{1}{6} + \frac{1}{18} = \boxed{\frac{2}{9}}. \end{aligned}$$

(c)

$$P(H\mid A=1) = \frac{P(H)P(A=1\mid H)}{P(A=1)} = \frac{(1/3)(2/3)}{1/3} = \boxed{\frac{2}{3}}.$$

(d)

$$P(H\mid A=1,B=1) = \frac{(1/3)(2/3)(3/4)}{2/9} = \frac{1/6}{2/9} = \boxed{\frac{3}{4}}.$$

(e)

$$P(B=1\mid A=1) = \frac{P(A=1,B=1)}{P(A=1)} = \frac{2/9}{1/3} = \boxed{\frac{2}{3}}.$$

(f) A and B are **not independent** without conditioning on the class, because

$$P(A=1,B=1) = \frac{2}{9} \quad \text{but} \quad P(A=1)P(B=1) = \frac{1}{3}\cdot\frac{7}{12} = \frac{7}{36}.$$

Since $\frac{2}{9} = \frac{8}{36} \neq \frac{7}{36}$, the required equality fails.

Teaching points for the reflection questions

- Both active indicators provide evidence in favour of H, so observing both increases the posterior probability from $\frac{2}{3}$ to $\frac{3}{4}$.
- A and B are independent *within* each event class. Mixing the classes can introduce dependence in the overall population.
- The denominator in Bayes' theorem is the probability of the *observed evidence*, including contributions from all relevant classes.